Hidden Feature-Guided Semantic Segmentation Network for Remote Sensing Images

Zhen Wang, *Member, IEEE*, Shanwen Zhang, *Member, IEEE*, Chuanlei Zhang, *Member, IEEE*, and Buhong Wang, *Member, IEEE*

***Abstract*— For semantic segmentation of remote sensing images, convolutional neural networks (CNNs) have proven to be powerful tools. However, the existing CNN-based methods have the problems of feature information loss, serious interference by clutter information, and ignoring the correlation between different scale features. To solve these problems, this article proposes a novel hidden feature-guided semantic segmentation network (HFGNet) for remote sensing images, which achieves accurate semantic segmentation by hierarchically extracting and fusing valuable feature information. Specifically, the hidden feature extraction module (HFE-M) is introduced to suppress the salient feature representation to mine more valuable hidden features. Meanwhile, the multifeature interactive fusion module (MIF-M) establishes the correlation between different features to achieve hierarchical feature fusion. The multiscale feature calibration module (MSFC) is constructed to enhance the diversity and refinement representation of hierarchical fusion features. Besides, the local-channel attention mechanism (LCA-M) is designed to improve the feature perception capability of the object region and suppress background information interference. We conducted extensive experiments on the widely used ISPRS 2-D Semantic Labeling dataset and the 15-Class Gaofen Image dataset. Experimental results demonstrate that the proposed HFGNet has advantages over several state-of-the-art methods. The source code and models are available at https://github.com/darkseid-arch/RS-HFGNet.**

***Index Terms*— Attention mechanism, convolution neural networks (CNNs), hidden feature extraction, remote sensing image, semantic segmentation.**

I. Introduction

With the continuous development of satellite and remote sensing technology [1], massive remote sensing data can be easily obtained [2], [3]. As an essential means of remote sensing image intelligent interpretation, semantic segmentation can assign semantic category information to each pixel [4], and it is widely used in land surveying [5], environment monitoring [6], disaster assessment [7], and so on. However, the unbalanced distribution of ground object categories, the high similarity of boundary regions, and the difficulty extracting detailed feature information in remote sensing images have significantly challenged semantic segmentation.

Over the past few decades, methods based on craft feature extraction have become the primary means of solving remote sensing image semantic segmentation problems [8]. These methods use color histogram (CH) [9], scale-invariant feature transform (SIFT) [10], and histogram of oriented gradients (HOG) [11] to obtain low-level features, such as shape, color, and texture of remote sensing images for semantic segmentation [12]. In addition, many classical classification and clustering algorithms, such as K-means clustering [13], fuzzy C-means [14], region growth [15], and superpixels [16], are also widely used in semantic segmentation tasks. However, with the continuous improvement of remote sensing image resolution, traditional methods cannot accurately extract feature information of ground objects and rely heavily on professional knowledge and domain experience, resulting in ineptitude to accurately achieve semantic segmentation.

In recent years, the development of convolutional neural networks (CNNs) has significantly improved the accuracy of semantic segmentation. The classical fully convolution neural network (FCN) [4] that uses CNNs as feature extractors has displayed exemplary performance in semantic segmentation tasks. Inspired by the end-to-end FCN [4] framework, many novel semantic segmentation networks have been successively proposed. For example, U-Net [17] and SegNet [18] use the encoder to obtain multiscale feature information, and the decoder is used to restore the feature map resolution and output prediction results. DeepLabV3 [19] uses dilated convolution instead of standard convolution to obtain spatial global context feature information by expanding the convolution receptive field. ICNet [20] uses a cascade structure to upsample low-resolution features and performs fine-grained fusion with high-resolution features. Recently, the attention mechanism has been widely used for semantic segmentation due to its advantages in acquiring long-range context information. Based on the attention mechanism, DANet [21] and CBAM [22] use spatial and channel attention mechanisms (CAMs) to improve the feature representation capability and complete the state-of-the-art performance. In summary, the semantic segmentation method based on CNNs and attention

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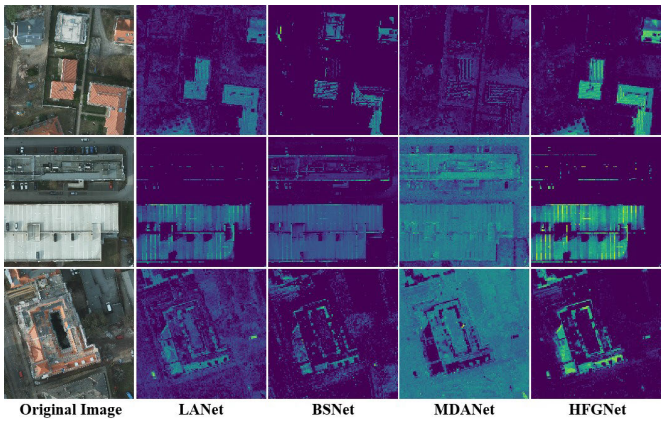


Fig. 1. Visual comparison of feature maps obtained by different methods, including LANet [23], BSNet [27], MDANet [30], and HFGNet.

mechanisms can achieve better semantic segmentation performance by obtaining multiscale and contextual feature information.

Currently, CNNs and attention mechanisms have also been widely used in remote sensing image semantic segmentation. Obtaining fine-grained feature information of large-scale remote sensing images is important for semantic segmentation tasks. Many feature extraction methods for remote sensing images have been proposed, such as LANet [23] that uses patch attention and attention embedding module to obtain local and global features. AERFC [24] obtains multiscale features by adaptively adjusting the receptive fields of different convolution kernels. AFNet [25] uses scale-feature attention and adjacent score map refinement module to enhance the saliency feature representations. MANet [26] introduces a region refinement module and multiscale collaborative learning to extract fine-grained features adaptively. BSNet [27] designs dynamic hybrid gradient convolution (DHGC) and coordinates sensitive attention to obtain multiscale boundary features. By using boundary attention and multitask learning, SBANet [28] can accurately extract deformation boundary features. SSAtNet [29] constructs pyramid attention pooling and pooling index correction modules to refine and calibrate multiscale features. MDANet [30] uses a sparse spatial sampling strategy and deformable attention to model the correlation of different feature information. However, these methods only focus on feature extraction of ground object regions, ignoring the importance of background region features. As shown in Fig. 1, we visualize the feature maps obtained by different methods. LANet [23] can effectively obtain local features, BSNet [27] has advantages in extracting boundary features, and MDANet [30] extracts more texture feature information, while our proposed HFGNet can enhance the saliency features of foreground regions via the guidance of background features.

Considering the shortcomings of remote sensing image semantic segmentation methods in feature extraction, we propose a hidden feature-guided semantic segmentation network (HFGNet) to extract different fine-grained features via hierarchical structure. Concretely, HFGNet includes four fundamental components: the hidden feature extraction module (HFE-M), the multifeature interactive fusion module (MIF-M), the multiscale feature calibration (MSFC), and local-CAM (LCA-M). To obtain more discriminative features, HFE-M uses saliency suppression to make the model mine more hidden feature information. MIF-M and MSFC interactively fuse features with different scales and enhance feature diversity representation. LCA-M can model the feature correlation of different channels, enhance the foreground feature representation, and suppress background information interference. In addition, we introduce adaptive weight parameters in the cross-entropy loss function to better optimize the semantic segmentation network.

The main contributions of this article are given as follows.

1) A novel HFGNet is proposed for remote sensing image segmentation. With the four constructed feature extraction and fusion components, i.e., HFE-M, MIF-M, MSFC, and LCA-M, HFGNet can adaptive obtain foreground region features and suppress background feature interference.
2) HFE-M and MIF-M can extract multiscale and fine-grained features in large-scale remote sensing images and realize feature fusion via feature interaction. MSFC and LCA-M perform calibration and channel correlation modeling on interactive fusion features, which can enhance the feature diversity of foreground regions.
3) To demonstrate the performance of the proposed semantic segmentation network, we perform a wide range of comparisons between different methods on ISPRS Postdam and Vaihingen datasets. The experiment results show that the proposed HFGNet is far more efficient than the other state-of-the-art methods.

This article is organized as follows. Section II presents related work. Section III illustrates the proposed HFGNet framework and related components. Section IV presents the experimental results and analysis. Finally, the conclusion is drawn in Section V.

II. Related Works

This section reviews related techniques widely used in semantic segmentation, including multiscale feature extraction, context feature extraction, and visual attention mechanism.

A. Multiscale Feature Extraction

The multiscale feature information of the region of interest (ROI) is essential for accurate semantic segmentation. Multiscale feature extraction of ground objects can solve large intraclass and small interclass differences in different scenes. The extract of multiscale feature information can solve large intraclass and small interclass differences in different scenes. In [31], the multiscale pyramid structure is used for feature extraction of different scale regions. MANet [32] uses dilated convolution and global average pooling (GAP) to construct a global context feature extraction module for multiscale feature extraction. Cue et al. [33] constructed the multipath polarimetric CNN (PolMPCNN) structure to obtain multiscale feature information of SAR images in complex scenes. In [34], the dynamic multiscale network (DMNet) composed of multiple dynamic convolution modules is used for different scale feature extraction. CcNet [35] combines

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group convolution, dilated convolution, and depthwise separable convolution to achieve multiscale feature extraction of different ground objects. DDGCN [36] combines the dynamic graph convolutional network and multidirectional FCN to obtain SAR image features of different scales and establish the correlation between different node features. Li et al. [37] designed a dilated-ResNet network (DRNet) using the multibranch residual structure and dilated convolution to extract multiscale features. SDFNet [38] uses the feature extraction module and the detail information extraction module to perform a fine-grained perception of boundary, shape, and contour features. LVSNet [39] designs the low-level feature extraction module and feature adaptive selection module to cascade and fuse different scale features. HRNet [40] uses an adaptive spatial pooling module and fine-grained feature perception module to enhance the representation of multiscale feature information.

The above method uses specially designed convolution and pooling to extract features from the foreground without considering the use of background information. Therefore, HFGNet uses HFE-M to suppress the feature representation of the foreground to obtain multiscale features.

B. Context Feature Extraction

The correlation of context feature information plays a vital role in the semantic segmentation of different scale regions. Extracting context features can model the correlation of pixels in different regions and make the semantic segmentation network more robust. For instance, HRNet [41] uses a dual attention mechanism and feature enhancement pyramid module to mine the correlation of context features for accurate semantic segmentation. MFENet [42] constructs spatial and edge extraction modules for primary feature extraction, and the context boost module is used to achieve feature enhancement. CPFNet [43] designs a scale-aware module for mining the correlation of local and global features considering the hierarchical of context features. In [44], the context encoder network (CE-Net) is used to model the correlation between different levels' context information. MFENet [45] uses a channel feature compression module to mine spatial context information and uses a multilevel feature aggregation module to filter redundant information. CGNet [46] uses the context-guided module to obtain the category correlation of different region pixels and uses a cross-fusion strategy to achieve context information aggregation. CTNet [47] combines spatial and channel context modules for fine-grained interaction of spatial and channel features to enhance the diversity of feature representation. CESNet [48] uses a feature encoder module to model the context and mine the correlation between local and global features.

However, these methods only extract context features and ignore feature diversity representation. Therefore, we construct MIF-M and feature calibration modules to extract global context information and enhance diversity representation.

C. Visual Attention Mechanism

The attention mechanism can enhance the saliency representation of valuable features and suppress redundant feature interference. Recently, many attention-based semantic segmentation networks have been proposed and achieved satisfactory performance. SCAttNet [49] combines channel and spatial attention mechanisms to establish the correlation between features and achieve fine-grained feature perception. MAResU-Net [50] uses a linear attention mechanism and skip connection for spatial multiscale feature extraction and multilevel feature interaction. Combining U-Net and CAM, Luo et al. [51] constructed an encoder–decoder structure semantic segmentation network for remote sensing images. RSANet [52] uses a regional self-attention mechanism to extract different scale features from different object regions and reduce redundant feature interference. In [53], the deformable attention pyramid module is used to adaptively obtain global semantic features, and different scale features are fused by the attention proximity module. Huang et al. [54] proposed an attention-guided label refinement network to enhance feature perception and aggregation capabilities for tiny object regions. To mine the correlation between local and global features, CANet [55] uses covariance attention to model the correlation of local–global information and enhance the feature diversity representation. HAssNet [56] combines channel, spatial, and position attention mechanisms to construct hybrid attention to extract features from different regions.

In summary, the use of an attention mechanism can enhance the object region features and suppress redundant feature interference. Therefore, we construct LCA-M for feature enhancement and correlation modeling.

III. Methodology

The overall architecture of HFGNet is shown in Fig. 2, which takes the original remote sensing image as input. The convolution kernels with different sizes (3×3, 5×5, and 7×7) are used to obtain the low-, middle-, and high-level feature information, respectively. The HFE-M suppresses the salient feature expression to obtain more discriminate feature information. The MIF-M establishes the correlation between different levels of hidden features to enhance the model's nonlinear learning ability. The multiscale feature calibration module (MSFC) enhances and refines the fusion features to improve the diversity representation of different scale features. The LCA-M is used to improve the feature perception ability of the object region and suppress background information interference. Overall, the proposed HFGNet is an end-to-end semantic segmentation network, which refines the representation of feature information via hierarchical processing to obtain accurate semantic segmentation results.

A. Hidden Feature Extraction Module

For remote sensing image semantic segmentation, convolution operation can obtain the salient features of the object region, but it ignores the role of background information. The extraction of background features can suppress the feature expression of the object region so that it can further mine valuable discriminative features. To suppress the salient representation of the object region features, we construct

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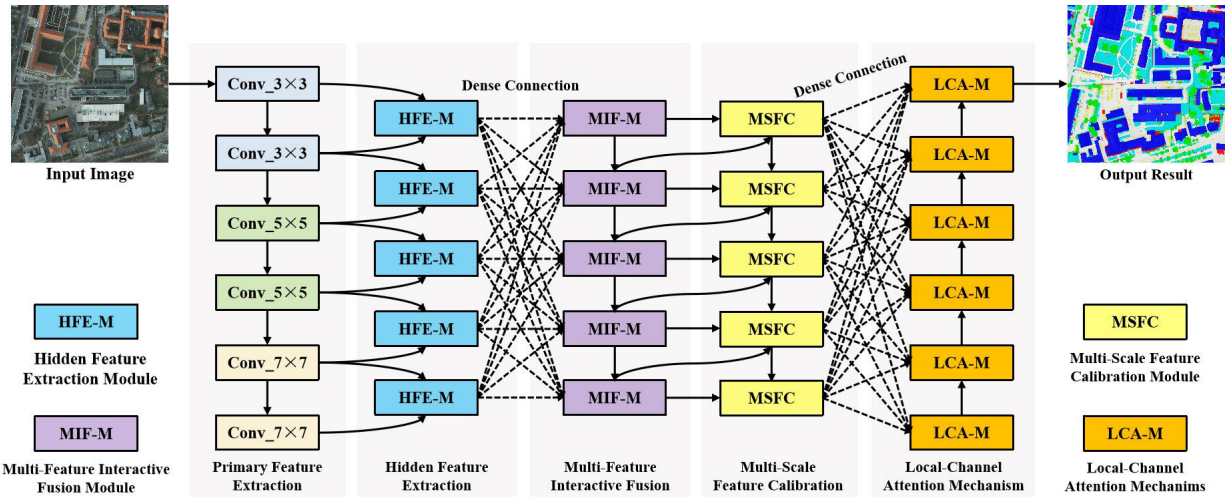


Fig. 2. Overall architecture of proposed HFGNet, including the HFE-M, MIF-M, MSFC, and LCA-M.

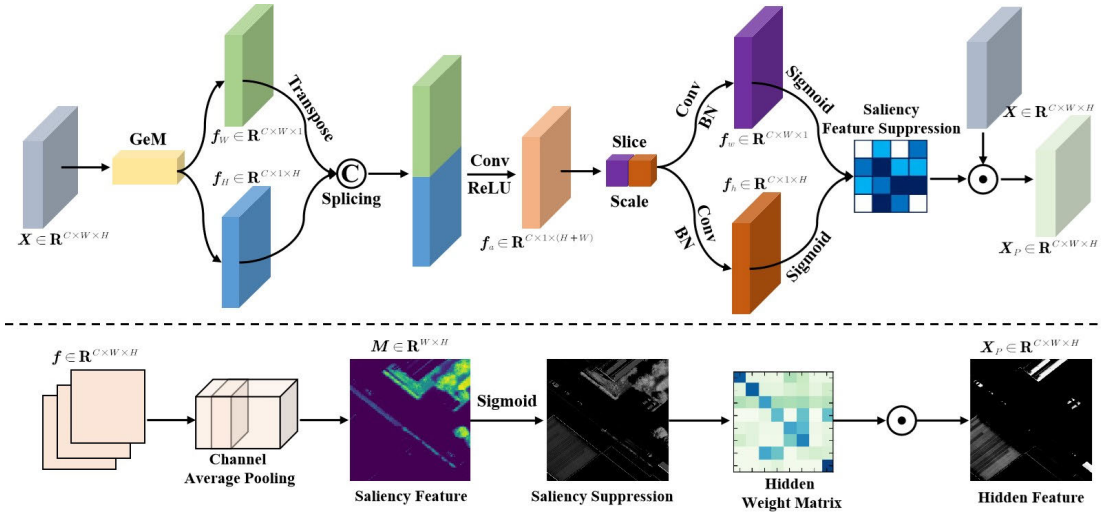


Fig. 3. Structure of HFE-M.

an HFE-M, as shown in Fig. 3. Assume that the input feature is $\boldsymbol{X} \in \mathbf{R}^{C\times W\times H}$, where C, W, and H represent the channel, width, and height of the feature map. Inspired by the coordinate attention mechanism [57], the HFE-M performs identical mapping on the input features to aggregate the features in the W and H directions of the spatial dimension to generate direction-aware feature maps. To obtain accurate spatial dimension direction feature information, the generalized mean (GeM) pooling [58] is used to preprocess feature $\boldsymbol{X}$. Specifically, we use two spatial extents of GeM pooling kernels $(H, 1)$ or $(1, W)$ to encode each channel along the horizontal coordinate and the vertical coordinate, respectively. The specific calculation process is given as follows:

$$\phi_{\mathrm{GeM}} = \left(1/\boldsymbol{X}(k) \sum_{x\in X(k)} x^{p(k)}\right)^{1/p(k)} \tag{1}$$

$$\boldsymbol{f}_H = \phi_{\mathrm{GeM_H}}(\boldsymbol{X}) \in \mathrm{R}^{C\times 1\times H} \tag{2}$$

$$\boldsymbol{f}_W = \phi_{\mathrm{GeM_W}}(\boldsymbol{X}) \in \mathrm{R}^{C\times W\times 1} \tag{3}$$

where ϕ_{GeM} represents the GeM pooling function; $\phi_{\mathrm{GeM_H}}$ and $\phi_{\mathrm{GeM_W}}$ represent the GeM pooling operation in the horizontal and vertical directions, respectively; $p(k)$ represents the hyperparameter corresponding to the kth channel; $p(k) = 1$ means global max pooling; $p(k) \to \infty$ means global max pooling; and $\boldsymbol{f}_H$ and $\boldsymbol{f}_W$ represent the salient features in H- and W-directions. Then, the stitching aggregation operation is performed on the directional features. To ensure consistent dimensions, we first transpose $\boldsymbol{f}_W$ and use convolution, activation, and batch normalization to mine the correlation of direction features. The specific calculation is given as follows:

$$\boldsymbol{f}_a = r\left(\eta\left(\zeta\left(\left[\boldsymbol{f}_H, \boldsymbol{f}_W\right]\right)\right)\right) \in \boldsymbol{R}^{C\times 1\times (H+W)} \tag{4}$$

where $\boldsymbol{f}_a$ is the mixed feature obtained by splicing spatial features in different directions. r and η represent the ReLU function and batch normalization; $\zeta(\cdot)$ represents the 1×1 convolution function for cross-channel information interaction; and $[\cdot]$ represents feature splicing on spatial dimension. The obtained mixed feature $\boldsymbol{f}_a$ is sliced and scaled in the channel dimension, and the specific calculation is given as follows:

$$\boldsymbol{f}_h, \boldsymbol{f}_w = s\left(\boldsymbol{f}_a\right) \tag{5}$$

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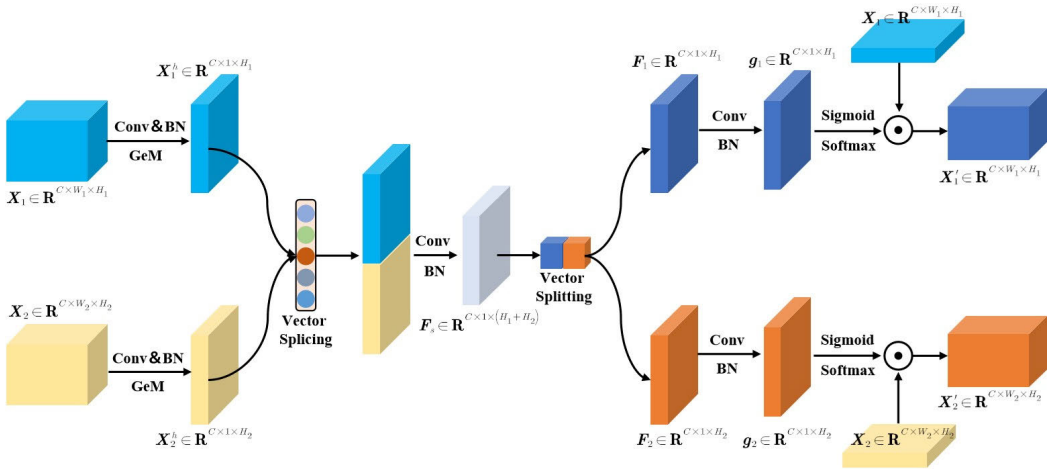


Fig. 4. Structure of MIF-M.

where $\boldsymbol{f}_h \in \mathbf{R}^{C\times 1\times H}$ and $\boldsymbol{f}_w \in \mathbf{R}^{C\times W\times 1}$ represent the feature tensors in the H- and W-directions on the channel dimensions, and s represents the slice function. To obtain weight features in different directions, use 1×1 convolution to process the feature tensor and reduce channel dimensions

$$\boldsymbol{g}_h = \sigma\left(\eta\left(\zeta\left(\boldsymbol{f}_h\right)\right)\right) \in \mathbf{R}^{C\times 1\times H} \tag{6}$$

$$\boldsymbol{g}_w = \sigma\left(\eta\left(\zeta\left(\boldsymbol{f}_w\right)\right)\right) \in \mathbf{R}^{C\times W\times 1} \tag{7}$$

where $\boldsymbol{g}_h$ and $\boldsymbol{g}_w$ represent different directional weight features and σ represents the sigmoid activation function. To obtain hidden feature information, we suppress the saliency information of weight features $\boldsymbol{g}_h$ and $\boldsymbol{g}_w$ so that the model can obtain more valuable discriminative information by focusing on hidden features. As shown in Fig. 3, the feature suppression module first uses channel average pooling to compress the weight feature to obtain the weight parameter $\boldsymbol{M}$ and then maps it to the same size as the original feature $\boldsymbol{X}$. The specific calculation is given as follows:

$$\boldsymbol{M}_i = \frac{1}{C}\sum_{z}^{C} \boldsymbol{g}_i \in \mathbf{R}^{(H\times 1)or(1\times W)} \tag{8}$$

$$\boldsymbol{g}_i^m = p(\boldsymbol{M}_i) \in \mathbf{R}^{H\times W} \tag{9}$$

where $\boldsymbol{M}_i$ represents the weight parameter obtained by the pooling operation and p represents the size mapping function, that is, matrix multiplication operation. The attention weights obtained are $\boldsymbol{g}_h^m \in \mathbf{R}^{W\times H}$ and $\boldsymbol{g}_w^m \in \mathbf{R}^{W\times H}$, and the obtained attention features are saliency suppressed by setting a threshold. Specifically, the feature value with a strong response in the feature map is set as the threshold, the pixel in the feature map is higher than the threshold to be judged as the discriminative region, and the weakening factor β is introduced. Only when the input feature pixel is equal to the threshold, the weakening factor is used to replace the pixel value. The specific discriminant formula for salient feature suppression is given as follows:

$$\boldsymbol{g}_{(i,j)}^s = \begin{cases} \beta, & \text{if } \boldsymbol{g}_{(i,j)} = \max\left(\boldsymbol{g}_{h\&w}^m\right) \\ 1, & \text{else.} \end{cases} \tag{10}$$

By suppressing the saliency information in the feature map, the hidden weight feature $\boldsymbol{g}_h^s$ and $\boldsymbol{g}_w^s$ can be calculated to obtain the hidden feature $\boldsymbol{X}_p$. The specific calculation is given as follows:

$$\boldsymbol{X}_p = \boldsymbol{X} \odot \boldsymbol{g}_h^s \odot \boldsymbol{g}_w^s \in \mathbf{R}^{C\times W\times H} \tag{11}$$

where $\odot$ represents the matrix dot product operation and $\boldsymbol{X}_p$ represents hidden features.

B. Multifeature Interactive Fusion Module

The interactive fusion of different feature information can effectively enhance the diverse representation, thereby improving the nonlinear learning ability of the semantic segmentation network. To enhance the diversity representation of discriminative features, we propose an MIF-M, which enhances feature representation by interactive modeling and feature fusion of channels and spatial information between different features. As shown in Fig. 4, given input features $\boldsymbol{X}_1 \in \mathbf{R}^{C\times W_1\times H_1}$ and $\boldsymbol{X}_2 \in \mathbf{R}^{C\times W_2\times H_2}$, the MIF-M first uses 1×1 convolution to preprocess the features and then uses GeM pooling to perform aggregation transformation in the height direction of the feature spatial

$$\boldsymbol{X}_1^h = \phi(\sigma(\eta(\zeta(\boldsymbol{X}_1)))) \in \mathbf{R}^{C\times 1\times H_1} \tag{12}$$

$$\boldsymbol{X}_2^h = \phi(\sigma(\eta(\zeta(\boldsymbol{X}_2)))) \in \mathbf{R}^{C\times 1\times H_2} \tag{13}$$

where $\boldsymbol{X}_1^h$ and $\boldsymbol{X}_2^h$ represent the obtained parametric features and ϕ represents the GeM pooling function. The parametric features are spliced to obtain the aggregated features, which are input into 1×1 convolution for feature fusion. The specific calculation is given as follows:

$$\boldsymbol{F}_s = r\left(\zeta\left(\left[\boldsymbol{X}_1^h, \boldsymbol{X}_2^h\right]\right)\right) \in \mathbf{R}^{C\times 1\times (H_1+H_2)} \tag{14}$$

where $\boldsymbol{F}_s$ represents the fusion features. The feature $\boldsymbol{F}_s$ is first sliced in the channel dimension to obtain complementary interaction features $\boldsymbol{F}_1 \in \mathbf{R}^{C\times 1\times H_1}$ and $\boldsymbol{F}_2 \in \mathbf{R}^{C\times 1\times H_2}$; then, 1×1 convolution is used to obtain features $\boldsymbol{g}_1$ and $\boldsymbol{g}_2$, and the diversity feature weight matrix is calculated by the softmax function

$$\boldsymbol{\omega}_{ij} = \frac{\exp\left(\boldsymbol{g}_{ij}\right)}{\sum_{k=1}^{c}\exp\left(\boldsymbol{g}_{ik}\right)}. \tag{15}$$

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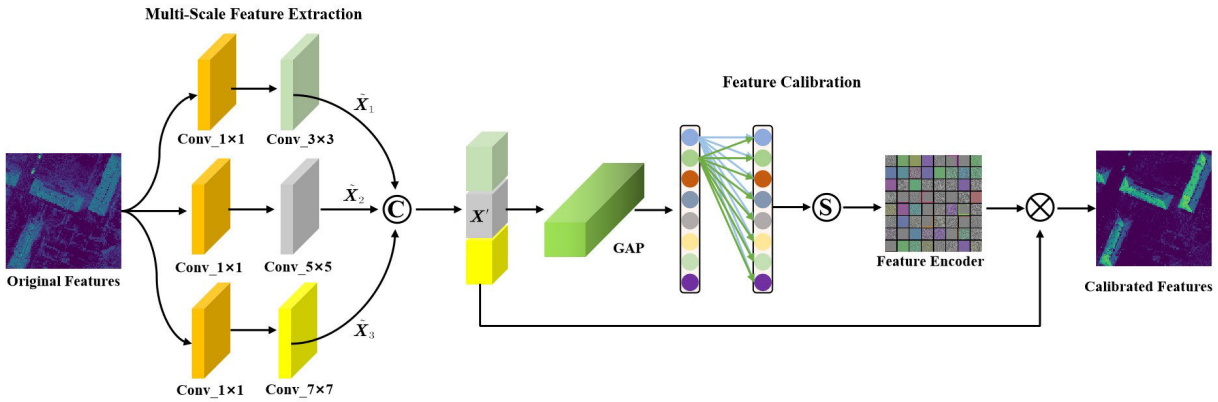


Fig. 5. Structure of MSFC.

The fusion feature is obtained by multiplying the diversity weight matrix with the original input feature

$$\boldsymbol{X}_1' = \boldsymbol{\omega}_1 \boldsymbol{X}_1 \in \mathbf{R}^{C \times W_1 \times H_1} \tag{16}$$

$$\boldsymbol{X}_2' = \boldsymbol{\omega}_2 \boldsymbol{X}_2 \in \mathbf{R}^{C \times W_2 \times H_2} \tag{17}$$

where $\boldsymbol{X}_1'$ and $\boldsymbol{X}_2'$ represent the fusion features of $\boldsymbol{X}_1$ and $\boldsymbol{X}_2$. Assuming that the input feature set is $I = \{\boldsymbol{X}_1, \boldsymbol{X}_2, \boldsymbol{X}_3\}$, the feature representation obtained by the interaction between feature $\boldsymbol{X}_i$ and other input features is given as follows:

$$\boldsymbol{Y}_i' = \sum_{\boldsymbol{X}_i \in I} \boldsymbol{X}_i' \tag{18}$$

where $\boldsymbol{X}_i'$ represents the interaction feature and $\boldsymbol{Y}_i'$ represents the diversity feature obtained by the addition operation. The obtained diversity feature $\boldsymbol{Y}_i'$ is fused with the input feature $\boldsymbol{X}_i$ to obtain the diversity fusion feature. The specific calculation is given as follows:

$$\boldsymbol{Y}_i = \boldsymbol{X}_i + \lambda \times \boldsymbol{Y}_i' \tag{19}$$

where λ is a hyperparameter adaptively adjusted during network training, and its initial value is set to 0.1.

C. Multiscale Feature Calibration Module

To further obtain the saliency feature representation of the ground object region and enhance the feature diversity expression, we propose an MSFC, which includes multiscale feature extraction and feature calibration. Specifically, we first design a parallel multiscale feature extraction structure consisting of different scale convolution kernels to obtain local and global feature information. Second, to make full use of the obtained feature information, we use the feature calibration mechanism composed of attention gate units to effectively filter and correct the feature information of different feature channels. The overall structure of multiscale feature calibration is shown in Fig. 5.

1) Multiscale Feature Extraction: For the input feature $\boldsymbol{X}$, three parallel convolution branches are used for feature extraction. Considering that the remote sensing image contains different scale object regions, we use convolution kernels with sizes of 3×3, 5×5, and 7×7 to extract multiscale features. To control the number of calculation parameters, the 1×1 convolution is introduced into each convolution branch for dimension reduction. Due to the large-scale variation of different object regions, we use instance normalization and LeakyReLU activate function to learn the appearance of statistical features. The calculation of multiscale feature extraction is given as follows:

$$\tilde{\boldsymbol{X}}_1 = \vartheta_{k=3}(\vartheta_{k=1}(\boldsymbol{X})) \tag{20}$$

$$\tilde{\boldsymbol{X}}_2 = \vartheta_{k=5}(\vartheta_{k=1}(\boldsymbol{X})) \tag{21}$$

$$\tilde{\boldsymbol{X}}_3 = \vartheta_{k=7}(\vartheta_{k=1}(\boldsymbol{X})) \tag{22}$$

where $\tilde{\boldsymbol{X}}_i$ represents the output of ith convolution branch, $\vartheta(\cdot)$ represents convolution, instance normalization, and LeakyReLU activate operations, and k represents the size of different convolution kernels.

2) Feature Calibration: To refine the fuse multiscale feature and reduce the impact of redundant background feature information, we use an attention gate unit to learn the weight information of different features and use the weight information to achieve feature calibration. The feature calibration steps are given as follows.

1) For the feature map obtained by different convolution branches, the fusion feature $\boldsymbol{X}' \in \mathbf{R}^{H \times W \times C}$ is obtained by feature concatenation on the channel dimension, where H, W, and C represent the height, width, and channel of the feature map. The calculation of feature concatenation is given as follows:

$$\boldsymbol{X}' = \text{concat}\left(\tilde{\boldsymbol{X}}_1, \tilde{\boldsymbol{X}}_2, \tilde{\boldsymbol{X}}_3\right). \tag{23}$$

2) To obtain the global feature information of $\boldsymbol{X}'$, the GAP is used to obtain the global statistical information of different channel features, which is represented as $x \in \mathbf{R}^C$. The cth element in x is calculated as follows:

$$x_c = F_{\text{GAP}}\left(\boldsymbol{X}_c'\right) = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \boldsymbol{X}_c'(i, j). \tag{24}$$

3) To enhance the correlation of different channels' feature information, we associate the feature information for the k channels adjacent to x_c and obtain channel weights by the interaction between different channels. The weight for the cth channel is calculated as follows:

$$w_c = \sigma\left(\sum_{j=1}^{k} w_c^j x_c^j\right), x_c^j \in \Omega_c^k \tag{25}$$

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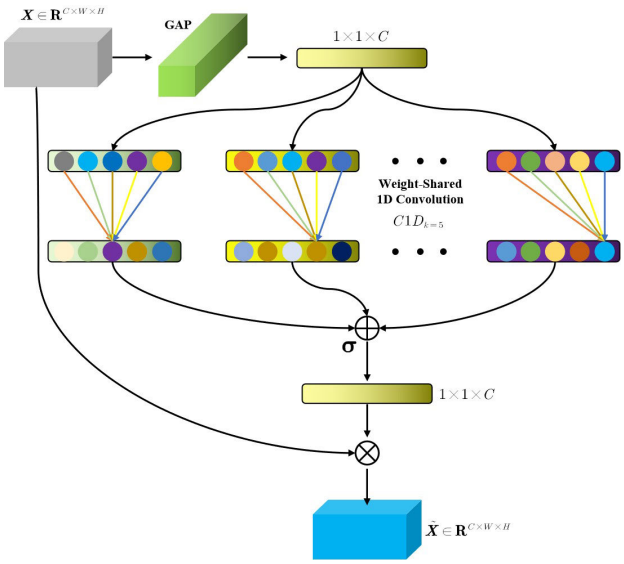


Fig. 6. Structure of LCA-M.

where σ represents the sigmoid function and Ω_c^k represents the set of k adjacent elements of x_c.

4) The obtained channel weight w is used to calibrate the feature map $\boldsymbol{X}'$, and the calculation process is given as follows:

$$\boldsymbol{Y} = w\boldsymbol{X}' = [w_1, \ldots, w_n, \ldots, w_c]\begin{bmatrix} \boldsymbol{X}'_1 \\ \vdots \\ \boldsymbol{X}'_n \\ \vdots \\ \boldsymbol{X}'_c \end{bmatrix} \tag{26}$$

where w_i represents the channel weight, $\boldsymbol{X}'_i$ represents the feature map to be calibrated, and $\boldsymbol{Y} \in \mathbf{R}^{H\times W\times C}$ is the feature calibration result.

D. Local-Channel Attention Mechanism

The correlation modeling of different channel features is beneficial to improve the feature representation ability. Although the existing CAM can establish the correlation between convolution channels, these methods cannot stably model channel correlations due to the characteristics of remote sensing images. To better model the channel feature correlation, we propose an LCA-M, which learns the weights of each channel by modeling the local correlation between channel information and uses deep Bayesian learning [59] to model the uncertainty of local channel information.

As shown in Fig. 6, the LCA-M uses GAP to generate feature vector $\boldsymbol{s}$ and uses 1-D fast convolution to model the local correlation between channels. Specifically, the 1-D convolution [60] is used to calculate the local correlation of channels on the feature vector, and the sigmoid function is used to obtain the weight of each channel. The calculation process is given as follows:

$$\omega = \sigma(C1D_k(\boldsymbol{s})) \tag{27}$$

where $C1D_k(\cdot)$ represents 1-D convolution with kernel size of k and σ represents the sigmoid function. To further model the uncertainty of different channel feature information, we introduce the Monte Carlo dropout operation in the LCA-M and use the approximate Bayesian inference [61] method for modeling uncertainty. Specifically, the learning parameter $\tilde{\mathbf{W}}$ is introduced into the LCA-M; for given input vector $\boldsymbol{s}$ and expected weight parameter $\tilde{\boldsymbol{s}}$, the distribution form that attention mechanism can obtain is given as follows:

$$p(\tilde{\mathbf{W}}|\boldsymbol{s}, \tilde{\boldsymbol{s}}). \tag{28}$$

To optimally (28), we use the Monte Carlo method to resample $\boldsymbol{s}$ under the condition of fewer calculation parameters, and the dropout operation is used to randomly perturb the resample features to approximate the posterior distribution $p(\tilde{\mathbf{W}}|\boldsymbol{s})$. The calculation process of generating channel feature vector $\tilde{\boldsymbol{s}}$ by Monte Carlo resample and dropout operation is given as follows:

$$\tilde{\boldsymbol{s}}_i = \mathcal{F}_{\text{dropout}}(s) = \text{Bernoulli}(p)\cdot s, \quad \text{for } i = 1, \ldots, n \tag{29}$$

where $\tilde{\boldsymbol{s}} = \{\tilde{\boldsymbol{s}}_1, \tilde{\boldsymbol{s}}_2, \ldots, \tilde{\boldsymbol{s}}_n\}$, n represents the number of 1-D convolution branches, p represents the probability of the dropout operation, and Bernoulli$(\cdot)$ represents the Bernoulli distribution. The LCA-M performs resampling and dropout operations for the aggregated channel features, and 1-D convolution with shared weight $\tilde{\mathbf{W}}$ is used to calculate the local correlation of feature vectors in different channels. The weight value ω is calculated as follows:

$$\omega = \sigma\left(\frac{1}{n}\sum_{1}^{n} C1D_k(\tilde{\boldsymbol{s}}_i)\right). \tag{30}$$

It can be seen from (30) that the uncertainty of the shared weight $\tilde{\mathbf{W}}$ can be modeled by performing multiple random resampling on the input features to obtain better regional feature extraction effect.

E. Adaptive Weight Loss Function

To better optimize the semantic segmentation network, based on cross-entropy loss and adaptive weight algorithm [62], we construct an adaptive weight loss function for the optimization of the semantic segmentation network. The cross-entropy loss can effectively evaluate the difference information between different probability distributions, that is, the information entropy difference between the prediction result and ground truth. The specific calculation is given as follows:

$$\mathcal{L} = -\frac{1}{n}(y_i(\ln a) + (1-y_i)\ln(1-a)) \tag{31}$$

$$\frac{\partial L}{\partial w_j} = \frac{1}{n}\sum_{x} x_j(\sigma(z) - y) \tag{32}$$

$$\frac{\partial L}{\partial b} = \frac{1}{n}\sum_{x} (\sigma(z) - y) \tag{33}$$

where x represents the input sample, y represents the ground truth, a represents the prediction result, and n represents the number of samples. $\sigma(z) - y$ is the difference between the predicted result and ground truth, where $z = wx + b$, and

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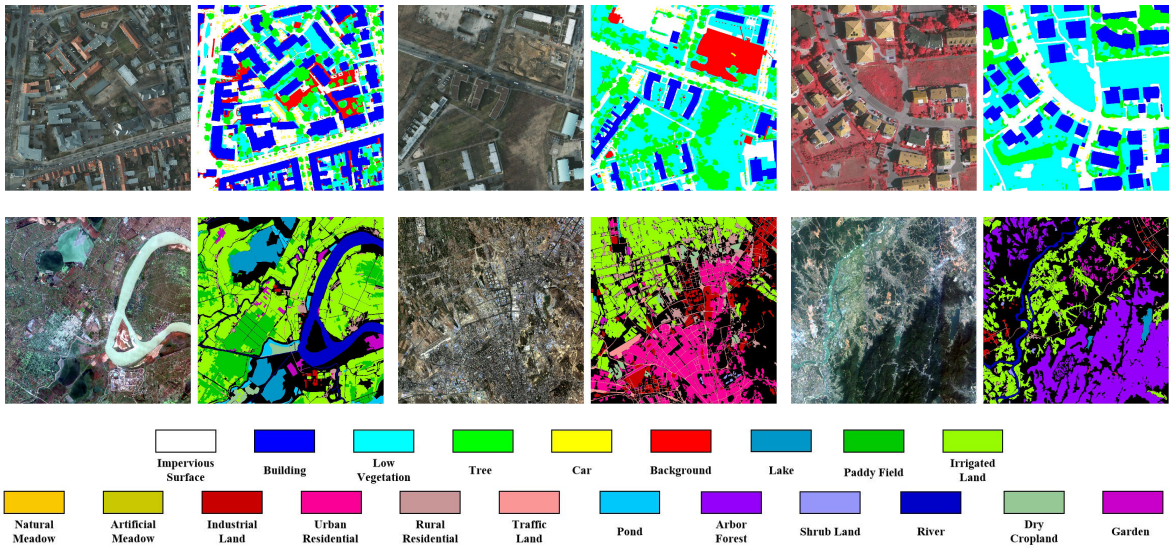


Fig. 7. Image sample and corresponding ground truth. The first row shows the Potsdam and Vahingen datasets, and the second row shows the GID-15 dataset.

the weight parameter w and bias b are adjusted by the calculated difference value. Since the class imbalance problem in remote sensing images significantly influences model training optimization, we introduce an adaptive weight algorithm in cross-entropy loss to solve the class imbalance problem by introducing adjustment parameters. The class weight function is defined as follows:

$$w_{\text{class}} = \frac{1}{\ln(\alpha + \beta P_{\text{class}})} \tag{34}$$

where w_{class} represents the weights of different classes, P_{class} represents the proportion of different sample classes, and α and β represent the hyperparameters adjusted according to the number of input sample classifications. Inspired by ENet [63], we set the α and β to 1.02 (i.e., restrict the class weights to be in the interval of [0, 50]). Based on (34), the function z in the cross-entropy loss is defined as $z = w_{\text{class}} + b$.

IV. Experiments

We evaluate the proposed semantic segmentation network HFGNet on ISPRS datasets (Postdam and Vaihingen) [64] and the 15-class Gaofen Image Dataset (GID-15) [65]. In this section, we first introduce the specific information of the dataset used and then give the model training details and algorithm quantitative evaluation metrics. In addition, we compare the proposed method with other semantic segmentation networks and perform reasonable ablation studies analysis.

A. Dataset Information

The Postdam, Vaihingen, and GID-15 datasets are used to verify the algorithm performance. Fig. 7 shows some samples and ground truth. As shown in Fig. 7, different ground object categories have high similarity and ambiguous boundaries.

1) Potsdam Dataset: The Postdam dataset contains six object categories of impervious surface, building, low vegetation, tree, car, and background. The dataset contains 38 remote sensing images, and we use 14 images as the training set and the rest as the test set. For each image, we only use near-infrared, red, and green band data. To facilitate the model training, we randomly crop the original image with a size of 6000 × 6.000 pixels into 140 patches with 512 × 512 pixels.

2) Vaihingen Dataset: The Vaihingen dataset contains the same six object categories as the Potsdam dataset. The difference is that the higher density of ground objects in the Vaihingen dataset makes it difficult to segment boundary regions. The dataset contains 33 remote sensing images of 2500 × 2.500 pixels, of which 17 images are used as the training set, while 16 images serve as the testing set. Similarly, we crop each original image as 24 patches of 512 × 512 pixels.

3) GID-15 Dataset: Different from the Potsdam and Vaihingen datasets, the GID-15 dataset contains 15 types of ground objects collected using the Gaofen-3 satellite. The remote sensing image resolution is 1-m panchromatic/4-m multispectral. The dataset provides a total of 150 remote sensing images with a size of 6800 × 7.200 pixels. We randomly crop the original image into multiple 512 × 512 pixel patches.

B. Training Details

1) Prevent Overfitting and Data Preprocessing: Limited by the number of samples, we use data augmentation and transfer learning during the model training process. Inspired by Penatti et al. [66], in all experiments, we use the ResNet (ResNet-50/ResNet-101) models as backbone networks and pretrained it on the ImageNet [67] dataset. To obtain refined features, the output stride of the backbone network is set to 12. Moreover, the multigrid method [68] is used in the backbone.

2) Configuration of Network: The experiments are running with Pytorch 1.11.0 and Python 3.7.0. All the experiments are performed on a workstation equipped with Intel i9-12900T CPU, 64-GB RAM, NVIDIA GTX GeForce 3090 GPU, and Ubuntu 18.04 operation system. In the optimization process, we set the initial learning rate to 0.001, and the stochastic gradient descent (SGD) with a momentum of 0.9 is used as the optimizer. According to Liu et al. [25], we adopt the poly learning strategy to automatically adjust the model learning rate. The initial iteration number of the model is set

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Algorithm 1 Training Framework for HFGNet Model

Input: High-resolution remote sensing image $\boldsymbol{I} \in \mathbf{R}^{H\times W\times C}$ and correspond ground truth $\boldsymbol{G}$.

Output: Prediction results of the testing set.

Set $batch_size = 8$, the weight attenuation of learnable parameters is 1×10^{-5}, the maximum iteration number is $n = 3.5 \times 10^4$, initial learning rate $r = 0.9$, optimizer SGD ($momentum = 0.9$), the stride $s = 12$ of backbone;
After data augmentation preprocessing, images with size of $512 \times 512 \times 3$ and corresponding labels are obtained;
Train the HFGNet semantic segmentation network;
for $i = 1$ to n **do**
    Extract low-level, middle-level, and high-level features from the backbone network (ResNet-50/ResNet-101);
    Input different level features to the HFE-M to obtain hidden feature $\boldsymbol{X}_p$;
    Transfer $\boldsymbol{X}_p$ to the MIF-M to obtain diversity fusion feature $\boldsymbol{Y}_i$;
    Transfer the fusion feature $\boldsymbol{Y}_i$ to the MSFC to get calibration feature $\boldsymbol{Y}$;
    The LCA-M is used to establish the correlation of feature $\boldsymbol{Y}$ and output the prediction results;
    Calculation the loss between prediction results and ground truth, and update the model weight parameters;
    Validate the weight performance on the testing set;
    When there is a higher mIoU or mF1, save the weight.
break
    Get the optimal training weight.
end for
Use testing set with the trained model to obtain semantic segmentation results.

to 35 000 iterations. In addition, we set the batch size to 8. The detailed training framework of the HFGNet model is shown in Algorithm 1.

C. Evaluate Metrics

The overall accuracy (OA), F1_score, mean F1_score, and mean intersection over union (mIoU) are used as evaluate metrics. We first define tp, fp, fn, and tn as true positives, false positives, false negatives, and true negatives. The OA is the correctly classified pixel in all pixels as

$$\mathrm{OA} = \frac{tp}{tp + fp + fn + tn}. \tag{35}$$

The F1_score is the geometric mean between the precision (P) and recall (R) of each class as

$$\mathrm{F1} = 2 \times \frac{P \times R}{P + R} \tag{36}$$

$$P = \frac{tp}{tp + fp}, \quad R = \frac{tp}{tp + fn}. \tag{37}$$

The mIoU is the mean of IoU, and the IoU is defined as

$$\mathrm{IoU}(P_i \cup G_i) = \frac{|P_i \cap G_i|}{|P_i \cup G_i|} \tag{38}$$

where P_i and G_i are the set of prediction pixels and ground-truth pixels for the ith class. $\cap$ and $\cup$ represent intersection and union operations, and $\|$ is the number of pixels in the set.

D. Comparison With State-of-the-Art Methods

To analyze the advantages of the proposed method, we conducted performance verification on three different test sets. The proposed method is compared with BSNet [27], SBANet [28], SSAtNet [29], MDANet [30], and SPANet [69] in the Potsdam dataset. For the Vaihingen dataset, the proposed method is compared with the semantic segmentation networks: LANet [23], AERFC [24], AFNet [25], MANet [26], and PEGNet [70]. For the GID-15 dataset, the proposed HFGNet is compared with HPS-Net [65], ST-UNet [71], WiCoNet [72], TCHNet [73], and MATNet [74]. Following the setting of the train set, the image size of the test set is 512×512 pixels. It is worth noting that the source code or the author provides all experimental results.

1) Results on Potsdam Dataset: To better present the superiority of the proposed method, we first briefly introduce the compared methods and clarify the disadvantages of these methods.

1) *BSNet [27]:* This network consists of DHGC and coordinate sensitive attention (CSA). DHGC uses vanilla convolution and gradient convolution to obtain boundary-sensitive feature information. CAS is used to establish the local correlation of different features and optimize detailed segmentation results. However, BSNet only considers obtaining boundary features and ignores the importance of global features. As shown in Table I, the OA, mean F1, and mIoU of BSNet reach 90.72%, 91.52%, and 77.51%, respectively. The results in Fig. 8 show that BSNet cannot accurately segment dense scene regions.
2) *SBANet [28]:* To obtain refined feature information, SBANet uses the boundary attention module (BA-module) to obtain fine-grained boundary features and uses adaptive weight multitask learning to efficiently aggregation different features. SBANet realizes the reasonable fusion of different feature information but ignores the mining of semantic features. As shown in Table I and Fig. 8, SBANet cannot achieve better segmentation results for large-area ground objects.
3) *SSAtNet [29]:* This network uses the pyramid attention pooling module to adaptively obtain the multiscale features and uses the pooling index correction module to restore the detailed features. However, limited by the receptive field range of traditional convolution operation, its feature extraction effect on tiny object areas is poor. Table I shows that the mF1 of SSAtNet is only 85.82%, and the results in Fig. 8 also show that SSAtNet has the problem of missegmentation.
4) *MDANet [30]:* To obtain local and global features, MDANet uses the deformable attention module to establish the long-range relationship between different features and uses the spatial sampling module to obtain spatial relative position information of different ground objects. Although MDANet can fully obtain spatial information, its feature extraction ability for boundary

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TABLE I
QUANTITATIVE COMPARISON WITH DIFFERENT METHODS ON THE POTSDAM TEST SET, WHERE THE BOLD FONT IS THE HIGHEST SCORE

Methods	Per-class F1_score (%)					Evaluate Metrics		
	Imp.surf.	Building	Low.veg.	Tree	Car	OA(%)	mF1(%)	mIoU(%)
BSNet [27]	92.42	95.61	86.83	88.12	94.63	90.72	91.52	77.51
SBANet [28]	93.83	98.06	88.97	89.48	94.71	92.80	93.01	79.36
SSAtNet [29]	89.76	92.53	79.58	86.75	80.51	87.96	85.82	75.38
MDANet [30]	96.73	98.24	91.24	96.15	86.53	96.74	93.57	89.57
SPANet [69]	91.61	94.90	87.09	88.85	94.08	89.73	91.31	84.15
HFGNet (ResNet-50)	95.87	98.12	92.05	96.48	93.54	97.32	95.21	91.24
HFGNet (ResNet-101)	**96.93**	**98.75**	**93.21**	**97.14**	**94.27**	**98.42**	**96.05**	**92.53**

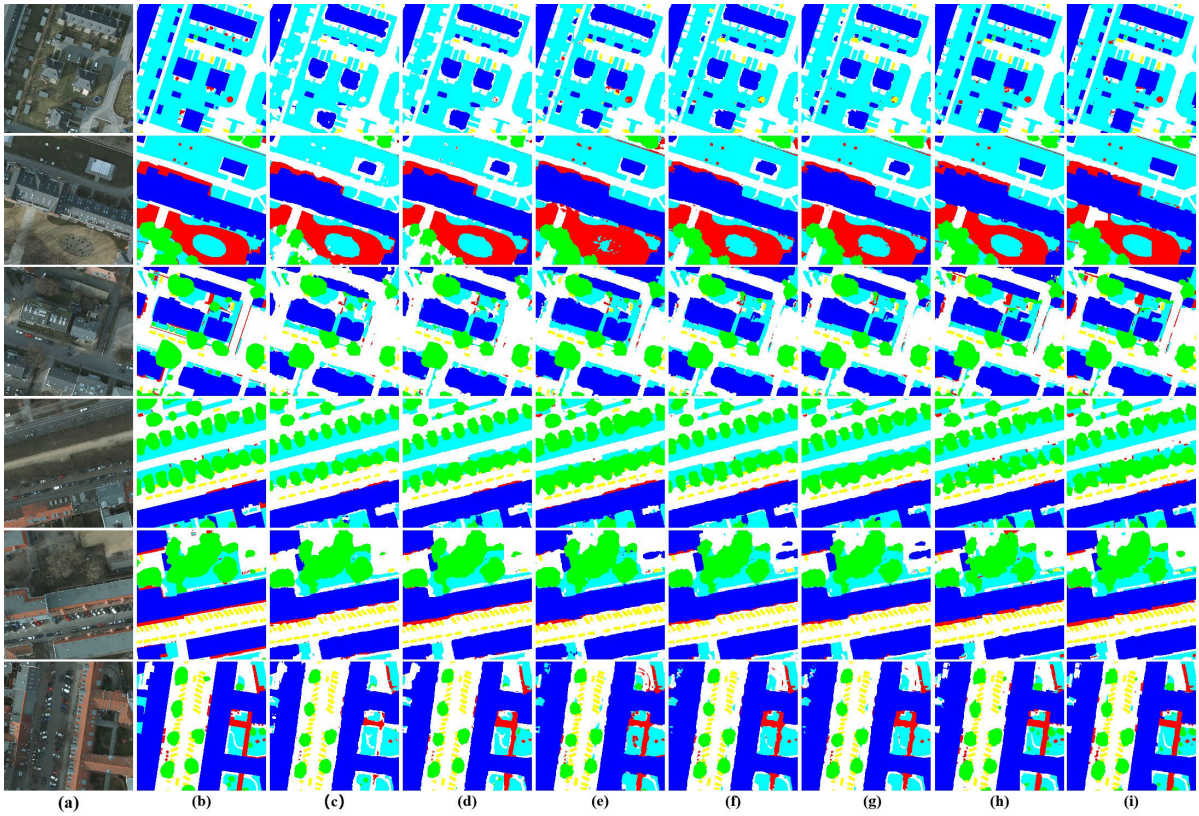


Fig. 8. Visualization comparison of segmentation maps of different methods on the Potsdam test set. (a) Original images. (b) Ground truth. (c) BSNet [27]. (d) SBANet [28]. (e) SSAtNet [29]. (f) MDANet [30]. (g) SPANet [69]. (h) HFGNet (RseNet-50). (i) HFGNet (ResNet-101).

details is poor, so there is still space for improvement. The results in Table I and Fig. 8 show that MDANet has lower segmentation accuracy for car objects.

5) *SPANet [69]:* This network consists of the successive pooling attention module (SPAM) and the feature fusion module (FFM). SPAM uses a parallel branch structure to obtain different feature information and perform deep feature mining. FFM is used to complement spatial and geometric features. However, SPANet is severely interfered with by background information, and Table I and Fig. 8 show the poor integrity of its segmentation results.

For our proposed HGFNet, as shown in Table I, it obtains the best segmentation results. For HGFNet using ResNet-50 as the backbone network, its OA, mF1, and mIoU reach 97.32%, 95.21%, and 91.24%, respectively. Compared with other methods, the proposed method shows great advantages. In addition, HGFNet using ResNet-101 as the backbone network further improves the segmentation accuracy. The results of Fig. 8 further prove that the proposed HFGNet can achieve accurate segmentation of different types of ground objects, and its segmentation results have better integrity.

2) Results on Vahingen Dataset: Different from the Potsdam dataset, the Vaihingen dataset contains many challenging scenarios. Table II and Fig. 9 show the quantitative and qualitative comparison of different methods on the Vaihingen test set. Next, we introduce the compared methods and analyze the advantages and drawbacks of these methods.

1) *LANet [23]:* To establish the connection between high-level semantic features and low-level pixel features. LANet uses a patch attention module to embed context information into pixel features and uses an attention

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TABLE II
QUANTITATIVE COMPARISON WITH DIFFERENT METHODS ON THE VAIHINGEN TEST SET, WHERE THE BOLD FONT IS THE HIGHEST SCORE

Methods	Per-class F1_score (%)					Evaluate Metrics		
	Imp.surf.	Building	Low.veg.	Tree	Car	OA(%)	mF1(%)	mIoU(%)
LANet [23]	94.90	92.41	82.89	88.92	81.31	89.83	88.09	81.35
AERFC [24]	93.14	95.93	85.96	90.35	88.53	92.03	90.78	83.62
AFNet [25]	93.47	95.91	86.02	90.73	87.24	91.65	90.67	83.17
MANet [26]	91.24	96.08	85.72	86.63	91.32	90.24	90.20	82.74
PEGNet [70]	91.72	95.43	81.06	88.92	87.24	89.53	88.87	81.56
HFGNet (ResNet-50)	94.38	97.24	91.15	95.34	92.40	95.87	94.10	90.83
HFGNet (ResNet-101)	**95.18**	**98.26**	**91.52**	**96.37**	**93.52**	**96.35**	**94.97**	**91.16**

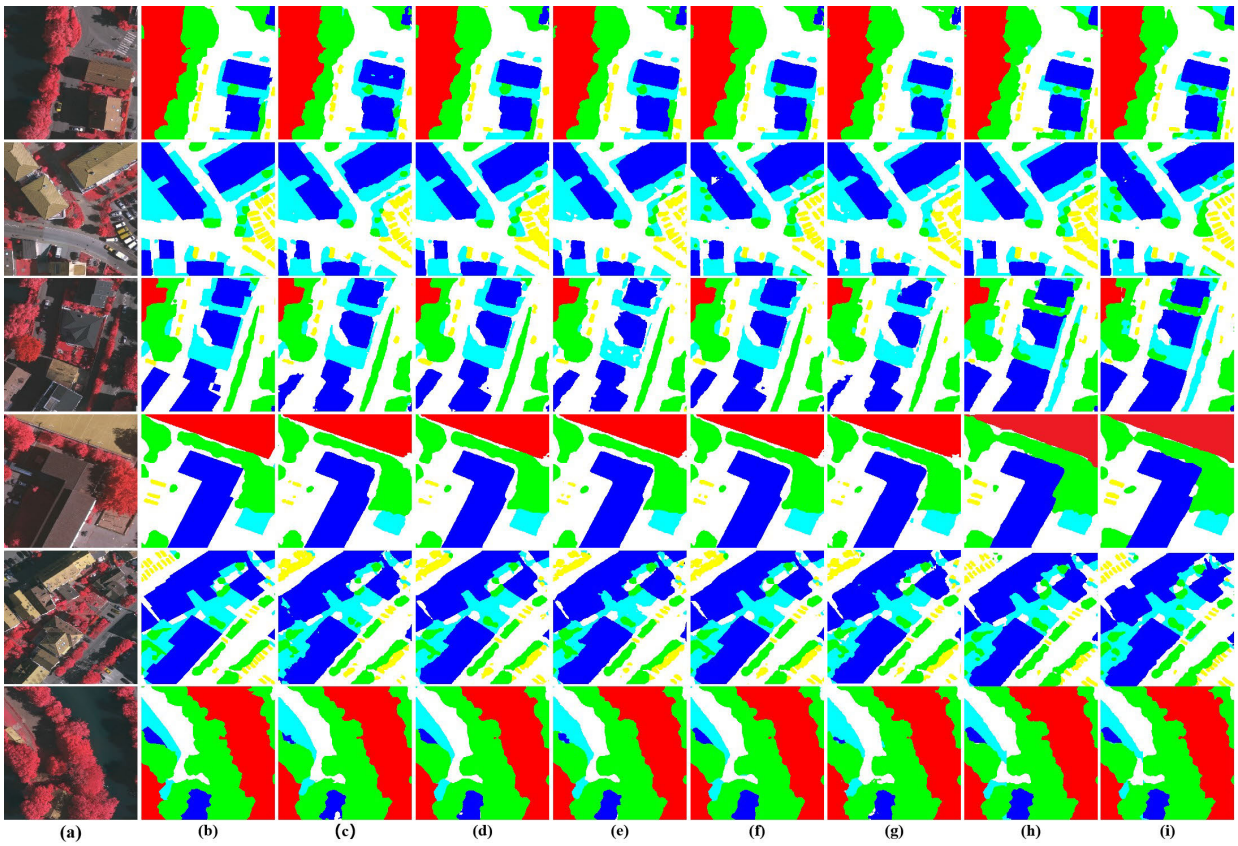


Fig. 9. Visualization comparison of segmentation maps of different methods on the Vaihingen test set. (a) Original images. (b) Ground truth. (c) LANet [23]. (d) AERFC [24]. (e) AFNet [25]. (f) MANet [26]. (g) PEGNet [70]. (h) HFGNet (RseNet-50). (i) HFGNet (ResNet-101).

embedding module to supplement local information of semantic features. LANet achieves the reasonable fusion of different features but ignores obtaining boundary features. As shown in Fig. 9, there is a boundary-missing phenomenon in the segmentation result.

2) *AERFC [24]:* To obtain the optimal convolution receptive field range, AERFC adaptively controls the convolution sampling location and automatically adjusts the effective receptive field range. In addition, AERFC uses spatial pyramid pooling to aggregate different scale features and mine context information. However, the results in Table II and Fig. 9 show that AERFC still has space for improvement, such as its poor segmentation results for ground objects in dense scenes.

3) *AFNet [25]:* The network consists of the scale-feature attention module (SFAM) and the scale-layer attention module (SLAM). SFAM is used to enhance the location and spatial features of different ground objects. SLAM is used to realize multiscale feature fusion and spatial dimension feature enhancement. However, AFNet cannot fully obtain the discriminative features, as shown in Fig. 9, which has serious false segmentation and is interfered with by the background information.

4) *MANet [26]:* The network uses interclass and intraclass region refinement for feature transformation and reduces feature information redundancy. Multiscale calibration learning is used to enhance the multiscale feature representation and suppress background information

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TABLE III
QUANTITATIVE COMPARISON WITH DIFFERENT METHODS ON THE GID-15 TEST SET, WHERE THE BOLD FONT IS THE HIGHEST SCORE

Methods	P.field	Irr.land	Dry cropl.	Garden	Arb.forest	Shr.land	Nat.mead.	Art.mead.	Ind.land
HPS-Net [65]	68.12	81.93	75.84	71.25	73.26	69.23	72.17	65.76	72.68
ST-UNet [71]	63.45	80.57	73.26	70.14	74.58	70.35	68.85	63.14	70.53
WiCoNet [72]	64.57	80.93	74.15	72.61	75.18	68.12	66.57	62.72	71.84
TCHNet [73]	70.28	83.52	76.28	73.52	75.46	72.41	71.50	68.05	73.24
MATNet [74]	71.60	84.73	77.41	71.85	76.92	70.05	74.16	67.32	74.32
HFGNet (ResNet-50)	74.54	87.13	84.39	77.64	80.05	76.87	**80.07**	75.53	**81.92**
HFGNet (ResNet-101)	**75.68**	**88.24**	**85.73**	**78.52**	**81.15**	**78.64**	79.83	**77.18**	80.43
Methods	Urb.resid.	Rur.resid.	Traff.land	River	Lake	Pond	OA(%)	mF1(%)	mIoU(%)
HPS-Net [65]	79.26	77.42	75.16	60.51	85.63	89.45	76.85	74.51	62.14
ST-UNet [71]	77.68	75.31	74.52	59.75	83.17	88.92	74.36	72.95	58.76
WiCoNet [72]	76.57	76.28	73.65	61.24	84.56	87.13	75.18	73.07	65.18
TCHNet [73]	80.29	78.57	78.95	64.12	88.25	90.15	78.92	76.31	68.25
MATNet [74]	81.38	80.42	79.30	65.36	87.06	89.26	79.13	76.74	70.36
HFGNet (ResNet-50)	86.70	87.83	86.12	69.55	92.04	**97.83**	85.67	82.55	78.95
HFGNet (ResNet-101)	**87.52**	**88.17**	**87.35**	**70.21**	**93.28**	97.42	**86.45**	**83.29**	**79.28**

interference. However, MANet ignores the correlation between different features. Table II and Fig. 9 also show that its segmentation results for small-scale objects still need to be improved.

5) *PEGNet [70]:* The network consists of the edge perception module (EPM), the high-level information module (HIM), and the feature aggregation module (FAM). EMP is used to obtain edge features and alleviate noise interference. HIM and FAM are used for semantic feature extraction and multiscale feature fusion. PEGNet achieves fine-grained semantic segmentation by a cascaded structure. However, the results of Fig. 9 show that PEGNet is less effective for segmenting adjacent regions due to its insufficient extraction of valuable features.

The results in Table II and Fig. 9 further demonstrate the advantages of HFGNet for remote sensing image semantic segmentation. HFGNet using ResNet-101 as the backbone network achieves the best results on the Vahingen test set. In addition, as shown in Fig. 9, HFGNet has a better segmentation effect and robustness, which can achieve accurate segmentation for different scales of ground objects.

3) Results on GID-15 Dataset: Since the GID-15 dataset contains a variety of ground object categories and complex scenes, it can further verify the robustness and advantages of the semantic segmentation network. The comparison results of different methods on the GID-15 test set are shown in Table III and Fig. 10. We first analyze the results of the comparison methods on the GID-15 test set.

1) *HPS-Net [65]:* To establish the relationship between features and pixels, HPS-Net uses multiple minibranch structures to obtain pixel- and feature-level information, and uses gated path selection and hidden path selection algorithms to mine deep features and optimize model parameters. However, HPS-Net is susceptible to background information in path selection. As shown in Table III and Fig. 10, HPS-Net has a poor segmentation effect on ground objects in complex scenes.

2) *ST-UNet [71]:* To fully obtain the context feature information, ST-UNet uses the spatial interaction module to establish the pixel-level correlation of different ground objects and uses the feature compression module to reduce the detailed feature information loss. In addition, the relational aggregation module is used for global and local feature fusion. As shown in Table III and Fig. 10, ST-UNet can obtain better segmentation results. However, ST-UNet ignores boundary feature extraction, so there is still space for improvement.

3) *WiCoNet [72]:* This network uses context branches to embed local features into global features, which can mine deep semantic feature information. WiCoNet fully considers the guidance of boundary features to the segmentation process and uses the local feature enhancement module to highlight the features of small-size object regions. However, it cannot obtain discriminative features in dense scenes. As shown in Fig. 10, WocoNet suffers from a false segmentation phenomenon.

4) *TCHNet [73]:* The network consists of encoder and decoder modules. The encoder module uses the Swin transformer to obtain long-range dependencies of multiscale features. The decoder module uses the atrous spatial pyramid pooling and CAM to restore the feature map resolution and output the segmentation results. However, TCHNet has the problem of detail feature loss in the process of feature fusion. As shown in Table III and Fig. 10, there is still space for improvement in the segmentation of small-scale objects.

5) *MATNet [74]:* To achieve fine-grained fusion of low-level and high-level features, MATNet constructs the kernel attention mechanism (KAM) and CAM. KAM is used to extract the nonlinear features and enhance the feature representation for different ground objects. CAM is used to enhance channel feature information and perform correlation modeling. As shown in Table III and Fig. 10, MATNet obtains better segmentation

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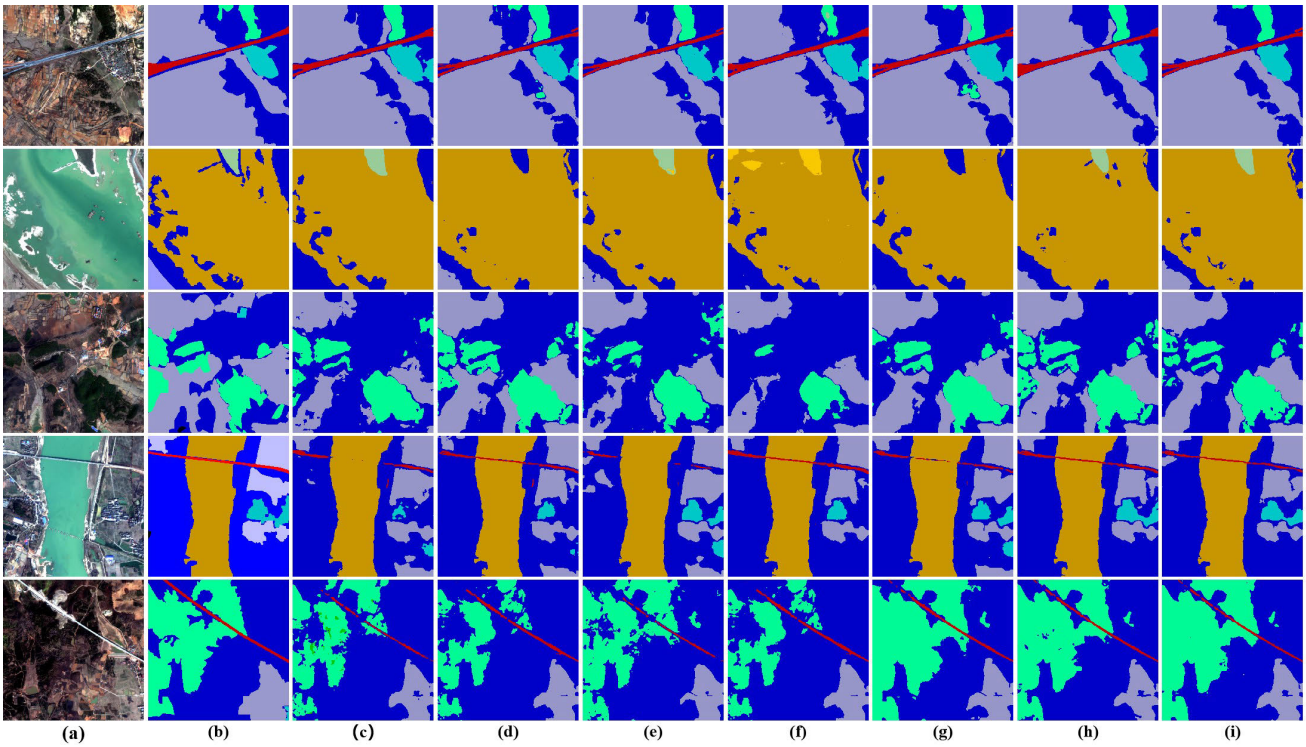


Fig. 10. Visualization comparison of segmentation maps of different methods on the GID-15 test set. (a) Original images. (b) Ground truth. (c) HPS-Net [62]. (d) ST-UNet [68]. (e) WiCoNet [69]. (f) TCHNet [70]. (g) MATNet [71]. (h) HFGNet (RseNet-50). (i) HFGNet (ResNet-101).

TABLE IV
ABLATION STUDIES OF HFE-M, MIF-M, MSFC, AND LCA-M

Models	HFE-M	MIF-M	MSFC	LCA-M	OA(%)	mF1(%)	mIoU(%)
Baseline	-	-	-	-	79.36	81.24	68.13
	✓	-	-	-	85.43	87.14	73.82
	✓	✓	-	-	87.27	89.28	81.54
	✓	✓	✓	-	91.48	93.17	86.42
	✓	✓	✓	✓	95.73	96.85	91.26

results but still needs to be improved for the boundary regions.

As shown in Table III and Fig. 10, the experimental results on the GID-15 dataset demonstrate the robustness and generalization ability of the proposed semantic segmentation network. Compared with other methods, our HFGNet can achieve the best segmentation accuracy and is significantly superior to the existing various methods.

E. Ablation Study

The proposed HFGNet contains key components HFE-M, MIF-M, MSFC, and LCA-M. To ensure the contribution of each component to the final results, we further conduct experimental verification, as shown in Table IV. For each component, we not only performed quantitative analysis but also visually compared its feature extraction process and its impact on segmentation results. Since the proposed HFGNet is similar in structure to BiSeNet [75], we use BiSeNet as the baseline model.

1) *Effectiveness of HFE-M:* The function of HFE-M is to suppress the significant feature representation of the object region, so the model can obtain more valuable discriminative features. As shown in Table IV, compared with the baseline model, the OA, mF1, and MIoU of the model with HFE-M increased by 6.07%, 5.90%, and 5.69%, respectively. Fig. 11 shows the feature visualization results of the baseline model and HFE-M. We can observe that HFE-M obtains more discriminative features, such as contour, shape, and texture by mining hidden information.
2) *Effectiveness of MIF-M:* The use of MIF-M can complete the interaction fusion of different branch features and establish the correlation between features. As shown in Table IV, the use of MIF-M effectively improves the semantic segmentation accuracy, with OA, mF1, and MIoU reaching 87.27%, 89.28%, and 81.54%, respectively. Fig. 12 shows the semantic segmentation results with the introduction of MIF-M. It can be seen that the use of MIF-M can effectively improve the semantic segmentation effect of small-size ground objects.
3) *Effectiveness of MSFC:* Fully extracting multiscale features of ground objects can effectively improve semantic segmentation accuracy. MSFC uses the multiscale feature extraction module to obtain local and global

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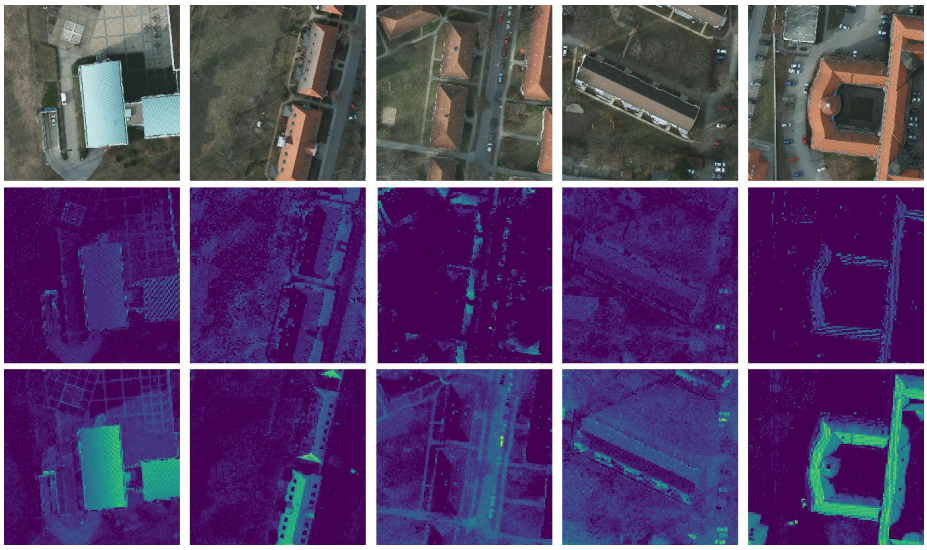
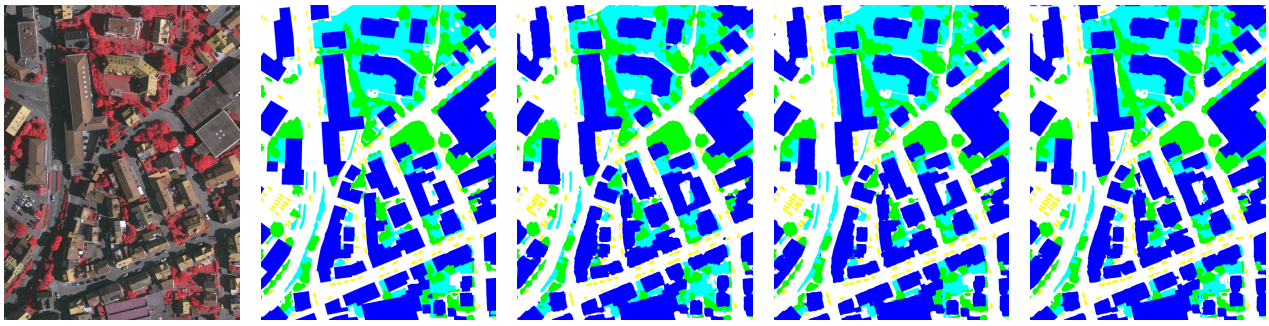
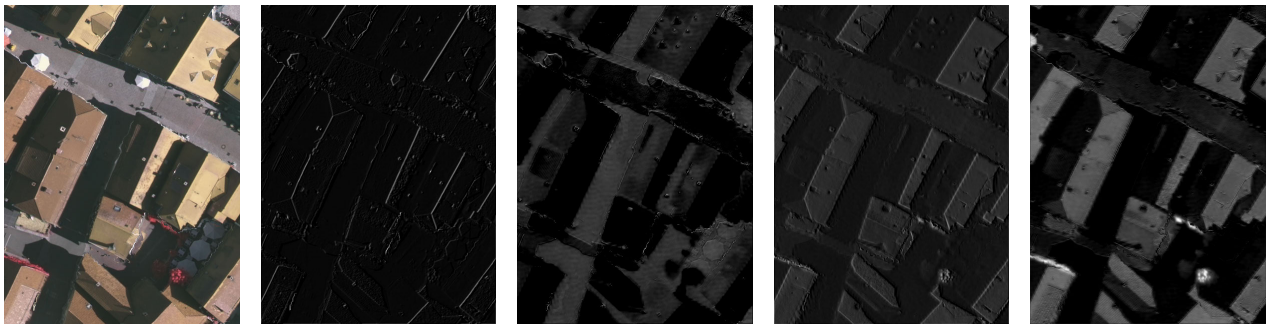


Fig. 11. Feature map comparison of the baseline model without and with HFE-M. The first row shows the original image, and the second and third rows show the feature maps generated by the baseline model and the baseline with HFE-M.



(a) (b) (c) (d) (e)

Fig. 12. Visual comparison of different model segmentation results. (a) Original images. (b) Ground truth. (c) Baseline model. (d) Baseline with HFE-M. (e) Baseline with HFE-M and MIF-M.



(a) (b) (c) (d) (e)

Fig. 13. Visual Comparison of different models' feature maps. (a) Original images. (b) Baseline model. (c) Baseline with HFE-M. (d) Baseline with HFE-M and MIF-M. (e) Baseline with HFE-M, MIF-M, and MSFC.

feature information, and uses the feature calibration unit to enhance the multiscale feature representation of the ground object region and suppress redundant feature interference. As shown in Table IV, using MSFC increased OA, mF1, and mIoU for the baseline combines HFE-M and MIF-M by 4.21%, 3.89%, and 4.88%, respectively. Fig. 13 shows the feature visual comparison of the baseline model and the MSFC. It can be seen that the use of MSFC highlights the salient feature representation of the ground object and filters the redundant feature interference.

4) *Effectiveness of LCA-M:* Modeling the feature correlation is conducive to improving the nonlinear learning ability of the model, making it focus on the feature extraction and representation of the object region, and suppressing the feature information interference of the nonobject region and background clutter. LCA-M enhances the feature diversity representation by modeling the correlation of local channel information. As shown in Table IV, the use of LCA-M further improves the semantic segmentation accuracy, with OA, mF1, and mIoU reaching 95.73%, 96.85%, and 91.26%,

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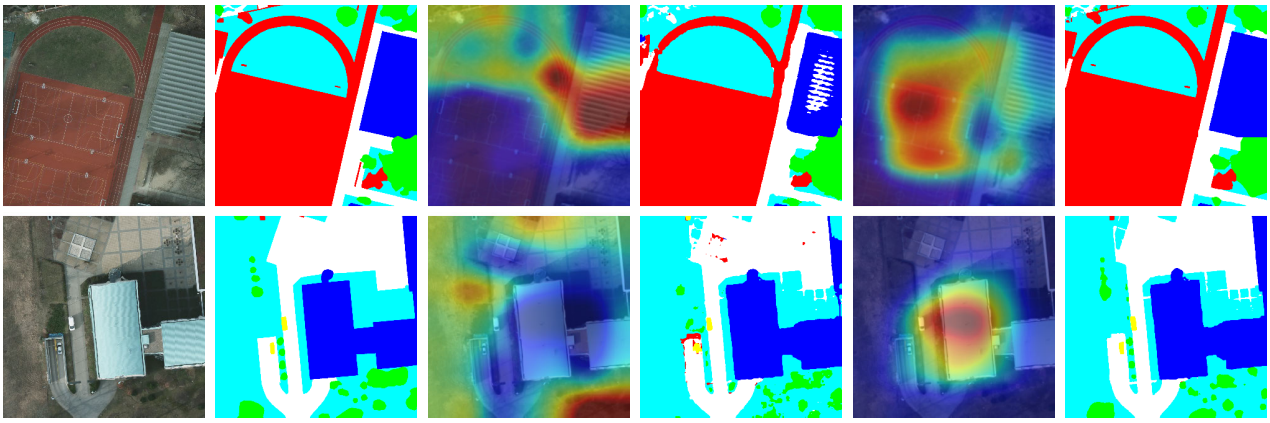


Fig. 14. Segmentation results of the baseline without and with LCA-M. (a) Original images. (b) Ground truth. (c) and (e) Visualization of corresponding attention feature maps. (d) and (f) Visualization of corresponding segmentation results.

respectively. Fig. 14 shows the visualization results of the attention feature map. Compared with the baseline model, LCA-M can focus on the feature extraction and representation of the object region and suppress the clutter feature information.

V. Conclusion

In this article, we propose a novel HFGNet to improve the semantic segmentation accuracy of remote sensing images. HFGNet contains four key components, i.e., HFE-M, MIF-M, MSFC, and LCA-M. By HFE-M, HFGNet can suppress the salient feature representation of the ground object region, allowing the model to learn more valuable feature information. By MIF-M, for multifeature interactive fusion, HFGNet can establish the correlation of different features and enhance the salient feature representation of ground objects. By MSFC, HFGNet achieves the calibration of multiscale feature information, enhances the feature diversity representation, and reduces the interference of redundant feature information. Furthermore, the LCA-M makes HFGNet establish the feature correlation of local channel information to obtain more valuable discriminative features. We verify the effectiveness and robustness of HFGNet through extensive experiments and achieve the SOTA performance. In addition, ablation studies demonstrate the effectiveness of each component in the proposed method. However, training perfect HFGNet still requires massive amounts of data and corresponding annotations. Therefore, introducing unsupervised or semisupervised learning methods in HFGNet is our future research direction.

References

[1] L. Ma, Y. Liu, X. Zhang, Y. Ye, G. Yin, and B. A. Johnson, "Deep learning in remote sensing applications: A meta-analysis and review," *ISPRS J. Photogramm. Remote Sens.*, vol. 152, pp. 166–177, Jun. 2019, doi: 10.1016/j.isprsjprs.2019.04.015.

[2] C. Toth and G. Jóźków, "Remote sensing platforms and sensors: A survey," *ISPRS J. Photogramm. Remote Sens.*, vol. 115, pp. 22–36, May 2016, doi: 10.1016/j.isprsjprs.2015.10.004.

[3] F. E. Fassnacht et al., "Review of studies on tree species classification from remotely sensed data," *Remote Sens. Environ.*, vol. 186, no. 2, pp. 64–87, Dec. 2016, doi: 10.1016/j.rse.2016.08.013.

[4] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017, doi: 10.1109/TPAMI.2016.2572683.

[5] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, no. 11, pp. 183–200, Aug. 2020, doi: 10.1016/j.isprsjprs.2020.06.003.

[6] M. Zhang, D. Zhu, W. Su, J. Huang, X. Zhang, and Z. Liu, "Harmonizing multi-source remote sensing images for summer corn growth monitoring," *Remote Sens.*, vol. 11, no. 11, pp. 1266–1273, May 2019, doi: 10.3390/rs11111266.

[7] S. T. Seydi, M. Akhoondzadeh, M. Amani, and S. Mahdavi, "Wildfire damage assessment over Australia using sentinel-2 imagery and MODIS land cover product within the Google earth engine cloud platform," *Remote Sens.*, vol. 13, no. 2, pp. 220–232, Jan. 2021, doi: 10.3390/rs13020220.

[8] L. Ma, M. Li, X. Ma, L. Cheng, P. Du, and Y. Liu, "A review of supervised object-based land-cover image classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 130, pp. 277–293, Aug. 2017, doi: 10.1016/j.isprsjprs.2017.06.001.

[9] M. J. Swain and D. H. Ballard, "Indexing via color histograms," in *Proc. 3rd Int. Conf. Comput. Vis.*, Dec. 1990, pp. 390–393, doi: 10.1109/ICCV.1990.139558.

[10] J. Cruz-Mota, I. Bogdanova, B. Paquier, M. Bierlaire, and J.-P. Thiran, "Scale invariant feature transform on the sphere: Theory and applications," *Int. J. Comput. Vis.*, vol. 98, no. 2, pp. 217–241, Jun. 2012, doi: 10.1007/s11263-011-0505-4.

[11] Z. Xiao, Q. Liu, G. Tang, and X. Zhai, "Elliptic Fourier transformation-based histograms of oriented gradients for rotationally invariant object detection in remote-sensing images," *Int. J. Remote Sens.*, vol. 36, no. 2, pp. 618–644, Feb. 2015, doi: 10.1080/01431161.2014.999881.

[12] D. Phiri and J. Morgenroth, "Developments in Landsat land cover classification methods: A Review," *Remote Sens.*, vol. 9, no. 9, pp. 967–975, Sep. 2017, doi: 10.3390/rs9090967.

[13] N. Dhanachandra, K. Manglem, and Y. J. Chanu, "Image segmentation using k-means clustering algorithm and subtractive clustering algorithm," *Proc. Comput. Sci.*, vol. 54, pp. 764–771, Jan. 2015, doi: 10.1016/j.procs.2015.06.090.

[14] W. Cai, S. Chen, and D. Zhang, "Fast and robust fuzzy C-means clustering algorithms incorporating local information for image segmentation," *Pattern Recognit.*, vol. 40, no. 3, pp. 825–838, Mar. 2007, doi: 10.1016/j.patcog.2006.07.011.

[15] F. Y. Shih and S. Cheng, "Automatic seeded region growing for color image segmentation," *Image Vis. Comput.*, vol. 23, no. 10, pp. 877–886, Sep. 2005, doi: 10.1016/j.imavis.2005.05.015.

[16] A. Levinshtein, A. Stere, K. N. Kutulakos, D. J. Fleet, S. J. Dickinson, and K. Siddiqi, "TurboPixels: Fast superpixels using geometric flows," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 31, no. 12, pp. 2290–2297, Dec. 2009, doi: 10.1109/TPAMI.2009.96.

[17] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351, Nov. 2015, pp. 234–241, doi: 10.1007/978-3-319-24574-4_28.

[18] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder–decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017, doi: 10.1109/TPAMI.2016.2644615.

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[19] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[20] H. Zhao, X. Qi, X. Shen, J. Shi, and J. Jia, "ICNet for real-time semantic segmentation on high-resolution images," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 405–420.

[21] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3141–3149, doi: 10.1109/CVPR.2019.00326.

[22] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 8–14.

[23] L. Ding, H. Tang, and L. Bruzzone, "LANet: Local attention embedding to improve the semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 426–435, Jan. 2021, doi: 10.1109/TGRS.2020.2994150.

[24] X. Chen et al., "Adaptive effective receptive field convolution for semantic segmentation of VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3532–3546, Apr. 2021, doi: 10.1109/TGRS.2020.3009143.

[25] R. Liu, L. Mi, and Z. Chen, "AFNet: Adaptive fusion network for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7871–7886, Sep. 2021, doi: 10.1109/TGRS.2020.3034123.

[26] P. He et al., "MANet: Multi-scale aware-relation network for semantic segmentation in aerial scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5624615, doi: 10.1109/TGRS.2022.3179379.

[27] J. Hou, Z. Guo, Y. Wu, W. Diao, and T. Xu, "BSNet: Dynamic hybrid gradient convolution based boundary-sensitive network for remote sensing image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5624022, doi: 10.1109/TGRS.2022.3176028.

[28] A. Li, L. Jiao, H. Zhu, L. Li, and F. Liu, "Multitask semantic boundary awareness network for remote sensing image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5400314, doi: 10.1109/TGRS.2021.3050885.

[29] Q. Zhao, J. Liu, Y. Li, and H. Zhang, "Semantic segmentation with attention mechanism for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5403913, doi: 10.1109/TGRS.2021.3085889.

[30] R. Zuo, G. Zhang, R. Zhang, and X. Jia, "A deformable attention network for high-resolution remote sensing images semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4406314, doi: 10.1109/TGRS.2021.3119537.

[31] K. Hu, M. Li, M. Xia, and H. Lin, "Multi-scale feature aggregation network for water area segmentation," *Remote Sens.*, vol. 14, no. 21, pp. 206–212, Jan. 2022, doi: 10.3390/rs14010206.

[32] R. Shang, J. Zhang, L. Jiao, Y. Li, N. Marturi, and R. Stolkin, "Multi-scale adaptive feature fusion network for semantic segmentation in remote sensing images," *Remote Sens.*, vol. 12, no. 5, pp. 872–881, Mar. 2020, doi: 10.3390/rs12050872.

[33] Y. Cui et al., "Polarimetric multipath convolutional neural network for PolSAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2021, doi: 10.1109/TGRS.2021.3071559.

[34] J. He, Z. Deng, and Y. Qiao, "Dynamic multi-scale filters for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 3561–3571, doi: 10.1109/ICCV.2019.00366.

[35] S. Feng, Z. Zhuo, D. Pan, and Q. Tian, "CcNet: A cross-connected convolutional network for segmenting retinal vessels using multi-scale features," *Neurocomputing*, vol. 392, pp. 268–276, Jun. 2020, doi: 10.1016/j.neucom.2018.10.098.

[36] F. Liu, X. Qian, L. Jiao, X. Zhang, L. Li, and Y. Cui, "Contrastive learning-based dual dynamic GCN for SAR image scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, May 20, 2022, doi: 10.1109/TNNLS.2022.3174873.

[37] J. Li, Z. L. Yu, Z. Gu, H. Liu, and Y. Li, "Dilated-inception net: Multi-scale feature aggregation for cardiac right ventricle segmentation," *IEEE Trans. Biomed. Eng.*, vol. 66, no. 12, pp. 3499–3508, Dec. 2019, doi: 10.1109/TBME.2019.2906667.

[38] G. Chen, Y. Dai, R. Li, Y. Zhao, L. Cui, and X. Yin, "SDFNet: Automatic segmentation of kidney ultrasound images using multi-scale low-level structural feature," *Expert Syst. Appl.*, vol. 185, pp. 115619–115632, Dec. 2021, doi: 10.1016/j.eswa.2021.115619.

[39] Q. Yan et al., "Attention-guided deep neural network with multi-scale feature fusion for liver vessel segmentation," *IEEE J. Biomed. Health Inform.*, vol. 25, no. 7, pp. 2629–2642, Jul. 2021, doi: 10.1109/JBHI.2020.3042069.

[40] J. Zhang, S. Lin, L. Ding, and L. Bruzzone, "Multi-scale context aggregation for semantic segmentation of remote sensing images," *Remote Sens.*, vol. 12, no. 4, pp. 701–715, Feb. 2020, doi: 10.3390/rs12040701.

[41] Z. Xu, W. Zhang, T. Zhang, and J. Li, "HRCNet: High-resolution context extraction network for semantic segmentation of remote sensing images," *Remote Sens.*, vol. 13, no. 1, pp. 71–82, Dec. 2020, doi: 10.3390/rs13010071.

[42] B. Zhang, W. Li, Y. Hui, J. Liu, and Y. Guan, "MFENet: Multi-level feature enhancement network for real-time semantic segmentation," *Neurocomputing*, vol. 393, pp. 54–65, Jun. 2020, doi: 10.1016/j.neucom.2020.02.019.

[43] S. Feng et al., "CPFNet: Context pyramid fusion network for medical image segmentation," *IEEE Trans. Med. Imag.*, vol. 39, no. 10, pp. 3008–3018, Oct. 2020, doi: 10.1109/TMI.2020.2983721.

[44] Z. Gu et al., "Ce-Net: Context encoder network for 2D medical image segmentation," *IEEE Trans. Med. Imag.*, vol. 38, no. 10, pp. 2281–2292, Oct. 2019, doi: 10.1109/TMI.2019.2903562.

[45] B. Chen, M. Xia, and J. Huang, "MFANet: A multi-level feature aggregation network for semantic segmentation of land cover," *Remote Sens.*, vol. 13, no. 4, pp. 731–742, Feb. 2021, doi: 10.3390/rs13040731.

[46] T. Wu, S. Tang, R. Zhang, J. Cao, and Y. Zhang, "CGNet: A light-weight context guided network for semantic segmentation," *IEEE Trans. Image Process.*, vol. 30, pp. 1169–1179, 2021, doi: 10.1109/TIP.2020.3042065.

[47] Z. Li, Y. Sun, L. Zhang, and J. Tang, "CTNet: Context-based tandem network for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 12, no. 2, pp. 9904–9917, Dec. 2021, doi: 10.1109/TPAMI.2021.3132068.

[48] H. Zhang et al., "Context encoding for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 7151–7160, doi: 10.1109/CVPR.2018.00747.

[49] H. Li, K. Qiu, L. Chen, X. Mei, L. Hong, and C. Tao, "SCAttNet: Semantic segmentation network with spatial and channel attention mechanism for high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 905–909, May 2021, doi: 10.1109/LGRS.2020.2988294.

[50] R. Li, S. Zheng, C. Duan, J. Su, and C. Zhang, "Multistage attention ResU-net for semantic segmentation of fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2021.3063381.

[51] H. Luo, C. Chen, L. Fang, X. Zhu, and L. Lu, "High-resolution aerial images semantic segmentation using deep fully convolutional network with channel attention mechanism," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 9, pp. 3492–3507, Sep. 2019, doi: 10.1109/JSTARS.2019.2930724.

[52] D. Zhao, C. Wang, Y. Gao, Z. Shi, and F. Xie, "Semantic segmentation of remote sensing image based on regional self-attention mechanism," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2021.3071624.

[53] F. Wu, F. Chen, X.-Y. Jing, C.-H. Hu, Q. Ge, and Y. Ji, "Dynamic attention network for semantic segmentation," *Neurocomputing*, vol. 384, pp. 182–191, Apr. 2020, doi: 10.1016/j.neucom.2019.12.042.

[54] J. Huang, X. Zhang, Y. Sun, and Q. Xin, "Attention-guided label refinement network for semantic segmentation of very high resolution aerial orthoimages," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4490–4503, Apr. 2021, doi: 10.1109/JSTARS.2021.3073935.

[55] Y. Liu, Y. Chen, P. Lasang, and Q. Sun, "Covariance attention for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 4, pp. 1805–1818, Apr. 2022, doi: 10.1109/TPAMI.2020.3026069.

[56] N. Lv et al., "A hybrid-attention semantic segmentation network for remote sensing interpretation in land-use surveillance," *Int. J. Mach. Learn. Cybern.*, vol. 12, no. 1, pp. 395–406, Jan. 2022, doi: 10.1007/s13042-022-01517-7.

[57] Q. Hou, D. Zhou, and J. Feng, "Coordinate attention for efficient mobile network design," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13708–13717, doi: 10.1109/CVPR46437.2021.01350.

[58] F. Radenović, G. Tolias, and O. Chum, "Fine-tuning CNN image retrieval with no human annotation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 7, pp. 1655–1668, Jul. 2019, doi: 10.1109/TPAMI.2018.2846566.

[59] H. Wang and D.-Y. Yeung, "A survey on Bayesian deep learning," *ACM Comput. Surv.*, vol. 53, no. 5, pp. 1–37, Sep. 2021, doi: 10.1145/3409383.

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[60] S. Kiranyaz, O. Avci, O. Abdeljaber, T. Ince, M. Gabbouj, and D. J. Inman, "1D convolutional neural networks and applications: A survey," *Mech. Syst. Signal Process.*, vol. 151, Apr. 2021, Art. no. 107398, doi: 10.1016/j.ymssp.2020.107398.

[61] P. Alquier, "Approximate Bayesian inference," *Entropy*, vol. 22, no. 11, pp. 1272–1278, Nov. 2020, doi: 10.3390/e22111272.

[62] F. Wang, Y. Li, H. Zhang, T. Hu, and X.-L. Shen, "An adaptive weight vector guided evolutionary algorithm for preference-based multi-objective optimization," *Swarm Evol. Comput.*, vol. 49, pp. 220–233, Sep. 2019, doi: 10.1016/j.swevo.2019.06.009.

[63] A. Paszke, A. Chaurasia, S. Kim, and E. Culurciello, "ENet: A deep neural network architecture for real-time semantic segmentation," 2016, *arXiv:1606.02147*.

[64] ISPRS. *International Society for Photogrammetry and Remote Sensing. 2D Semantic Labeling Challenge*. Accessed: Sep. 4, 2018. [Online]. Available http://www2.isprs.org/commissions/comm3/wg4/semantic-labeling.html

[65] K. Yang, X.-Y. Tong, G.-S. Xia, W. Shen, and L. Zhang, "Hidden path selection network for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5628115, doi: 10.1109/TGRS.2022.3197334.

[66] O. A. B. Penatti, K. Nogueira, and J. A. dos Santos, "Do deep features generalize from everyday objects to remote sensing and aerial scenes domains?" in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2015, pp. 44–51, doi: 10.1109/CVPRW.2015.7301382.

[67] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255, doi: 10.1109/CVPR.2009.5206848.

[68] G. Papandreou and P. Maragos, "Multigrid geometric active contour models," *IEEE Trans. Image Process.*, vol. 16, no. 1, pp. 229–240, Jan. 2007, doi: 10.1109/TIP.2006.884952.

[69] L. Sun, S. Cheng, Y. Zheng, Z. Wu, and J. Zhang, "SPANet: Successive pooling attention network for semantic segmentation of remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4045–4057, May 2022, doi: 10.1109/JSTARS.2022.3175191.

[70] S. Pan, Y. Tao, X. Chen, and Y. Chong, "Progressive guidance edge perception network for semantic segmentation of remote-sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2021.3058427.

[71] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408715, doi: 10.1109/TGRS.2022.3144165.

[72] L. Ding et al., "Looking outside the window: Wide-context transformer for the semantic segmentation of high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410313, doi: 10.1109/TGRS.2022.3168697.

[73] C. Zhang, W. Jiang, Y. Zhang, W. Wang, Q. Zhao, and C. Wang, "Transformer and CNN hybrid deep neural network for semantic segmentation of very-high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408820, doi: 10.1109/TGRS.2022.3144894.

[74] R. Li et al., "Multiattention network for semantic segmentation of fine-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607713, doi: 10.1109/TGRS.2021.3093977.

[75] C. Yu, J. Wang, C. Peng, C. Gao, G. Yu, and N. Sang, "BiSeNet: Bilateral segmentation network for real-time semantic segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 334–349.



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